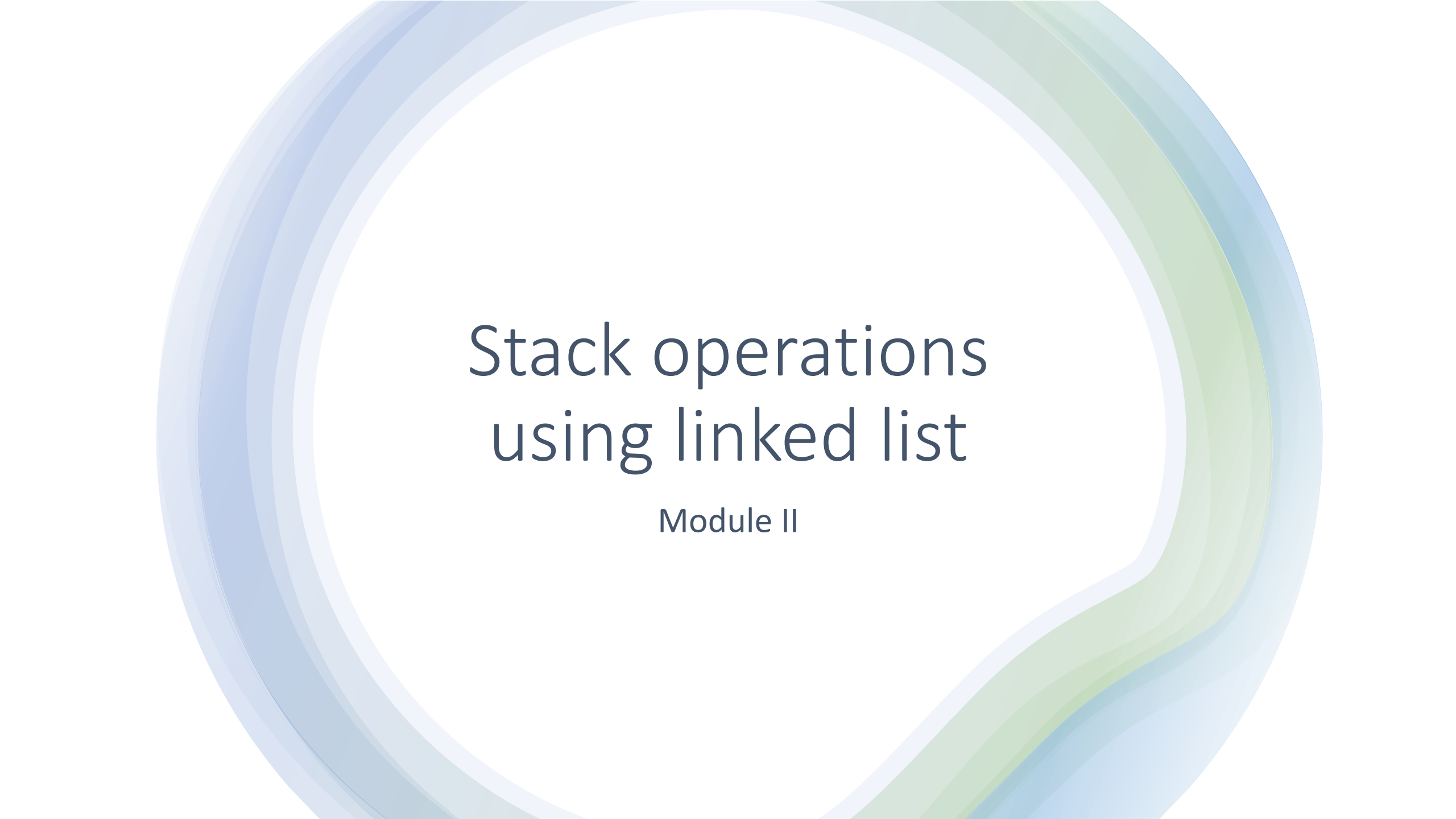




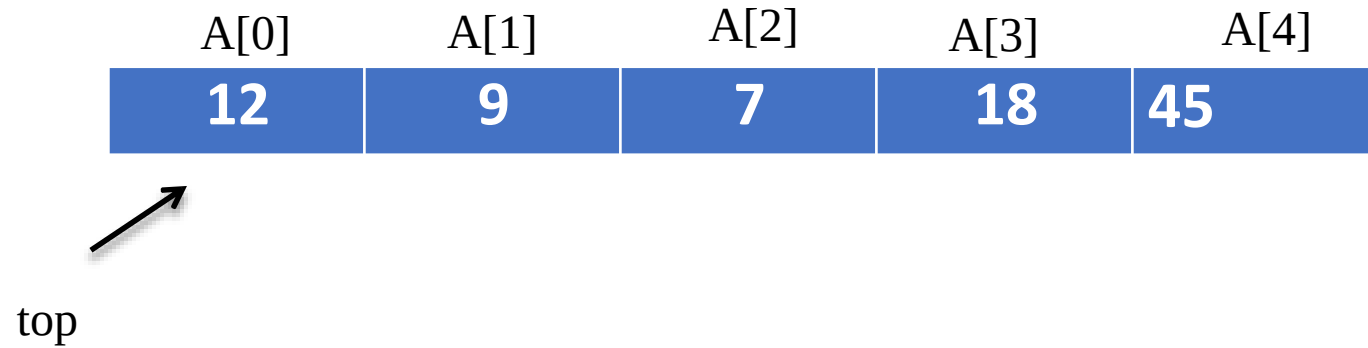
Stack operations using linked list

Module II

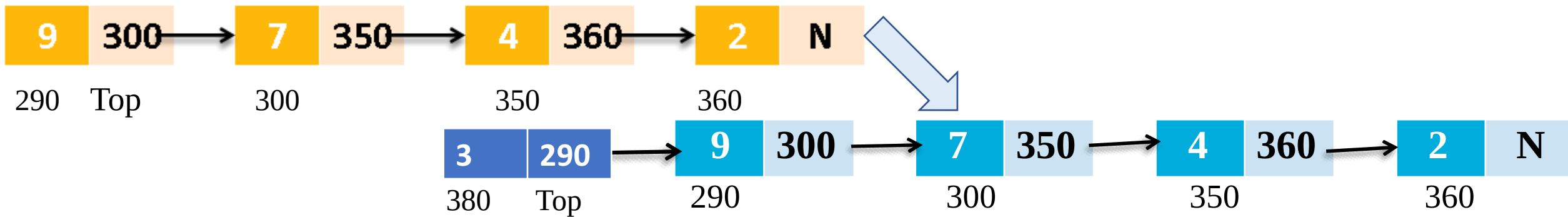
Representation of Stacks

- Stack data structure can be represented using arrays.
- Stack data structure can be represented using Linked List.

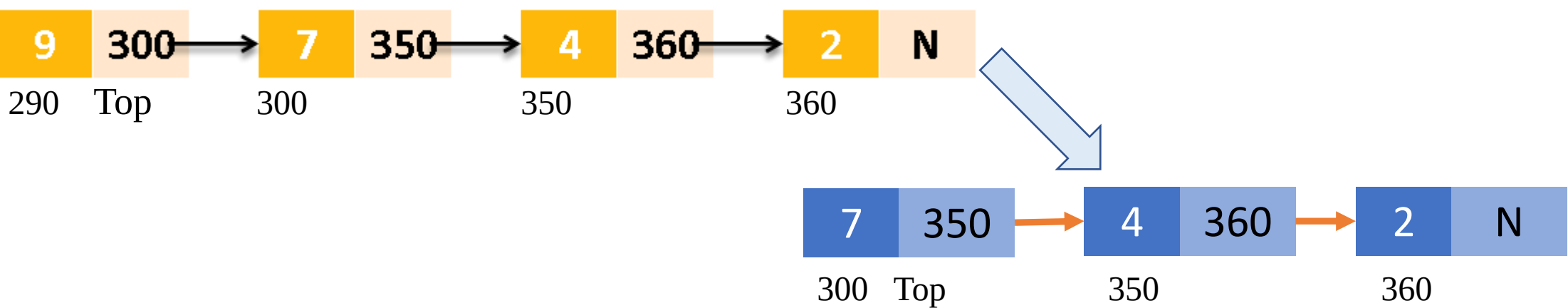
Array representation of Stack



Linked representation of Stack



Linked stack after inserting a node



Linked stack after deleting a node

Insert an element in stack using linked list

TOP=NULL

STEP-1: Allocate memory for the new node
& name it as NEW.

STEP-2: SET NEW → DATA = NUM
SET NEW → PTR = NULL

STEP-3: IF TOP = NULL
TOP=NEW

ELSE

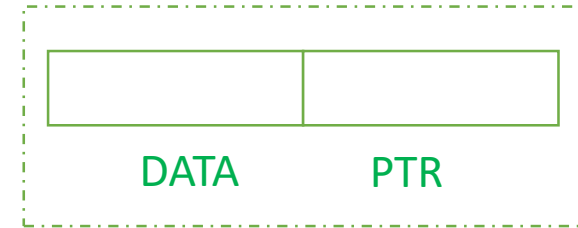
NEW → PTR = TOP

TOP=NEW

[END OF IF]

STEP-4: EXIT

NODE STRUCTURE



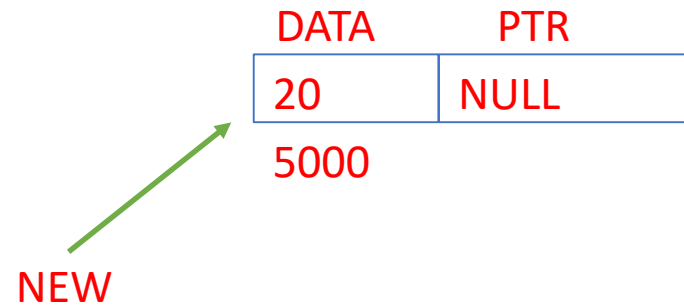
1. SINSERT_LINK(NUM)

STEP-1: Allocate memory for the new node & name it as NEW.

STEP-2: SET NEW → DATA = NUM

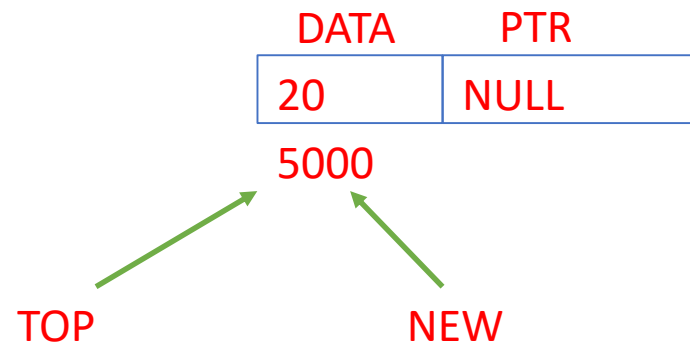
SET NEW → PTR = NULL

TOP=NULL



STEP-3: IF TOP = NULL
TOP=NEW
ELSE
NEW → PTR = TOP
TOP=NEW
[END OF IF]

TOP=NEW=5000

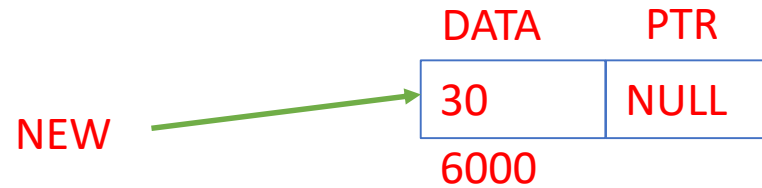


2. SINSERT_LINK(NUM)

STEP-1: Allocate memory for the new node & name it as NEW.

STEP-2: SET NEW → DATA = NUM

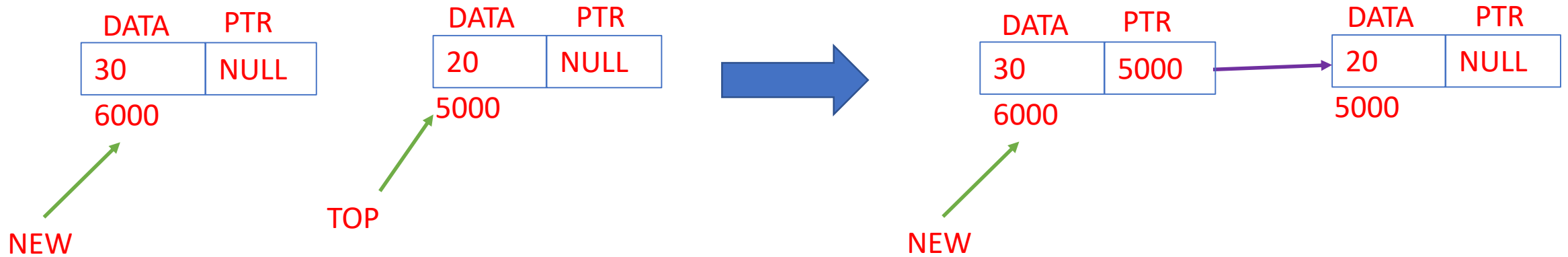
SET NEW → PTR = NULL



TOP=5000
NEW=6000

STEP-3: IF TOP = NULL
TOP=NEW
ELSE
NEW → PTR = TOP
TOP=NEW
[END OF IF]

TOP=6000
NEW=6000



3. SINSERT_LINK(NUM)

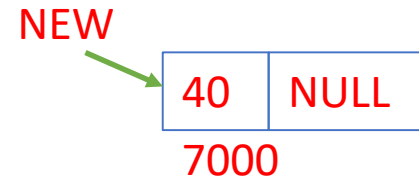
STEP-1: Allocate memory for the new node & name it as NEW.

STEP-2: SET NEW → DATA = NUM

SET NEW → PTR = NULL

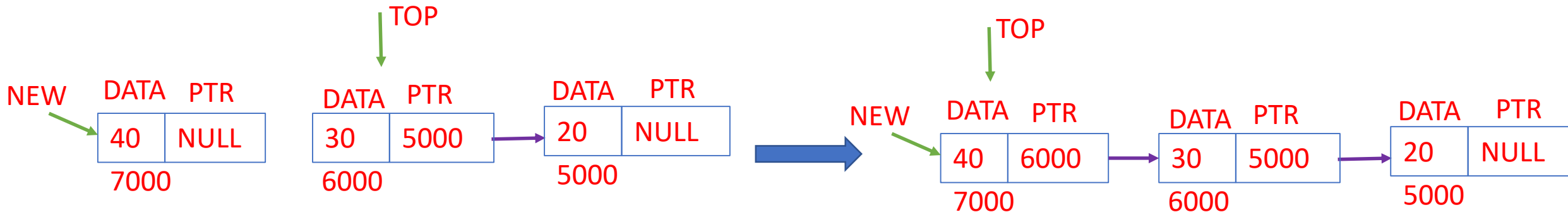
TOP=6000

NEW=7000

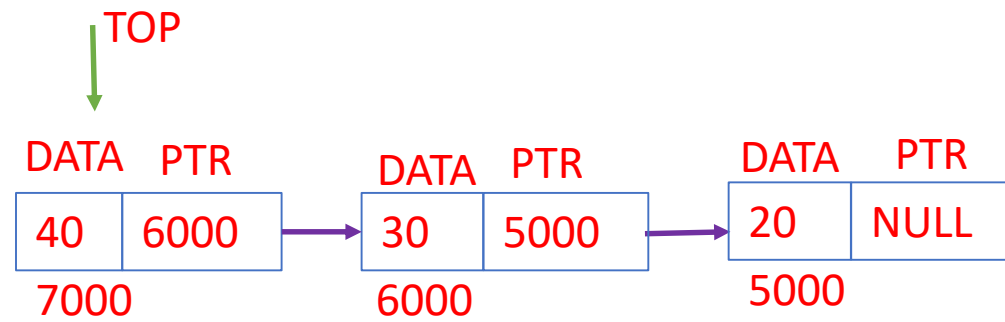


STEP-3: IF TOP = NULL
TOP=NEW
ELSE
NEW → PTR = TOP
TOP=NEW
[END OF IF]

TOP=7000
NEW=7000



TOP=7000



Delete an element in STACK using linked list

STEP-1: If TOP = NULL

 write “underflow”

 go to step 4.

 [End of if]

STEP-2: SET TEMP = TOP

 TOP=TOP→ PTR

STEP-3: FREE(TEMP)

STEP-4: EXIT

Delete an element in STACK using linked list

1. SDELETE_LINK()

STEP-1: If TOP = NULL

write "underflow"

go to step 4.

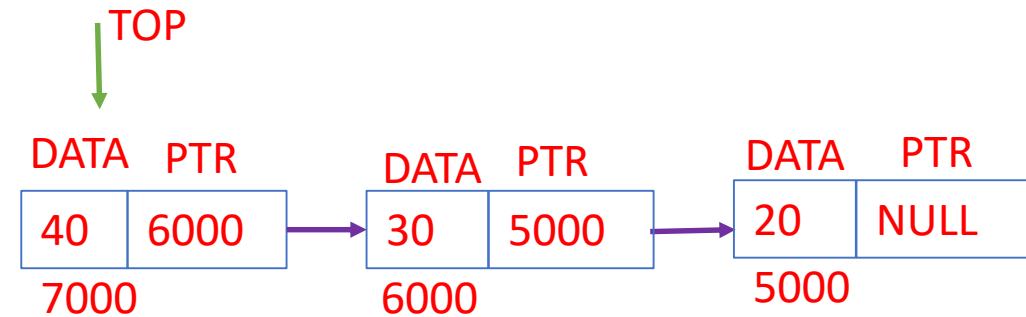
[End of if]

STEP-2: SET TEMP = TOP

TOP=TOP → PTR

STEP-3: FREE(TEMP)

STEP-4: EXIT



Delete an element in STACK using linked list

STEP-1: If TOP = NULL
write "underflow"
go to step 4.

[End of if]

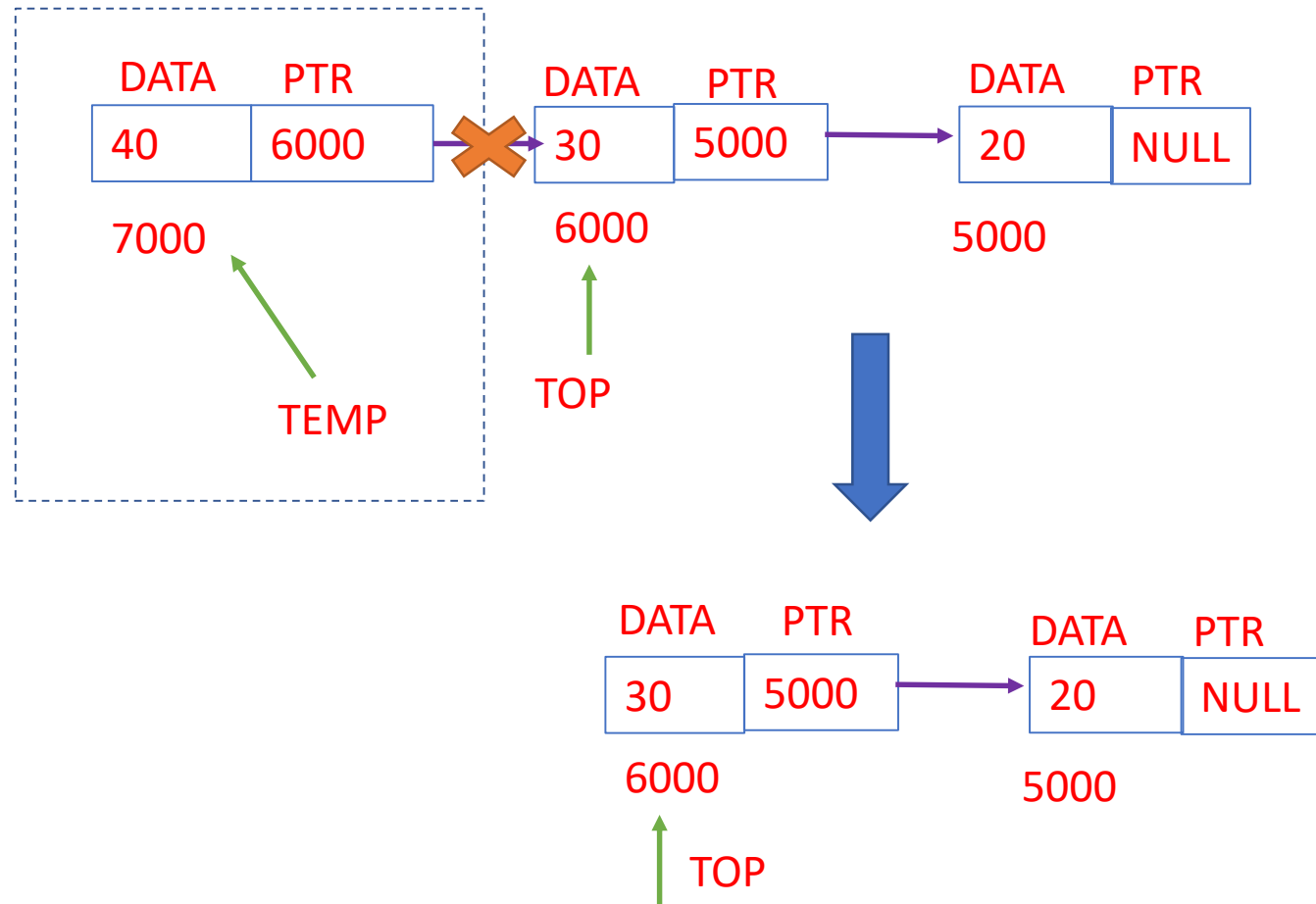
STEP-2: SET TEMP = TOP

TOP=TOP→ PTR

STEP-3: FREE(TEMP)

STEP-4: EXIT

1. SDELETE_LINK()



Delete an element in STACK using linked list

STEP-1: If TOP = NULL
write "underflow"
go to step 4.

[End of if]

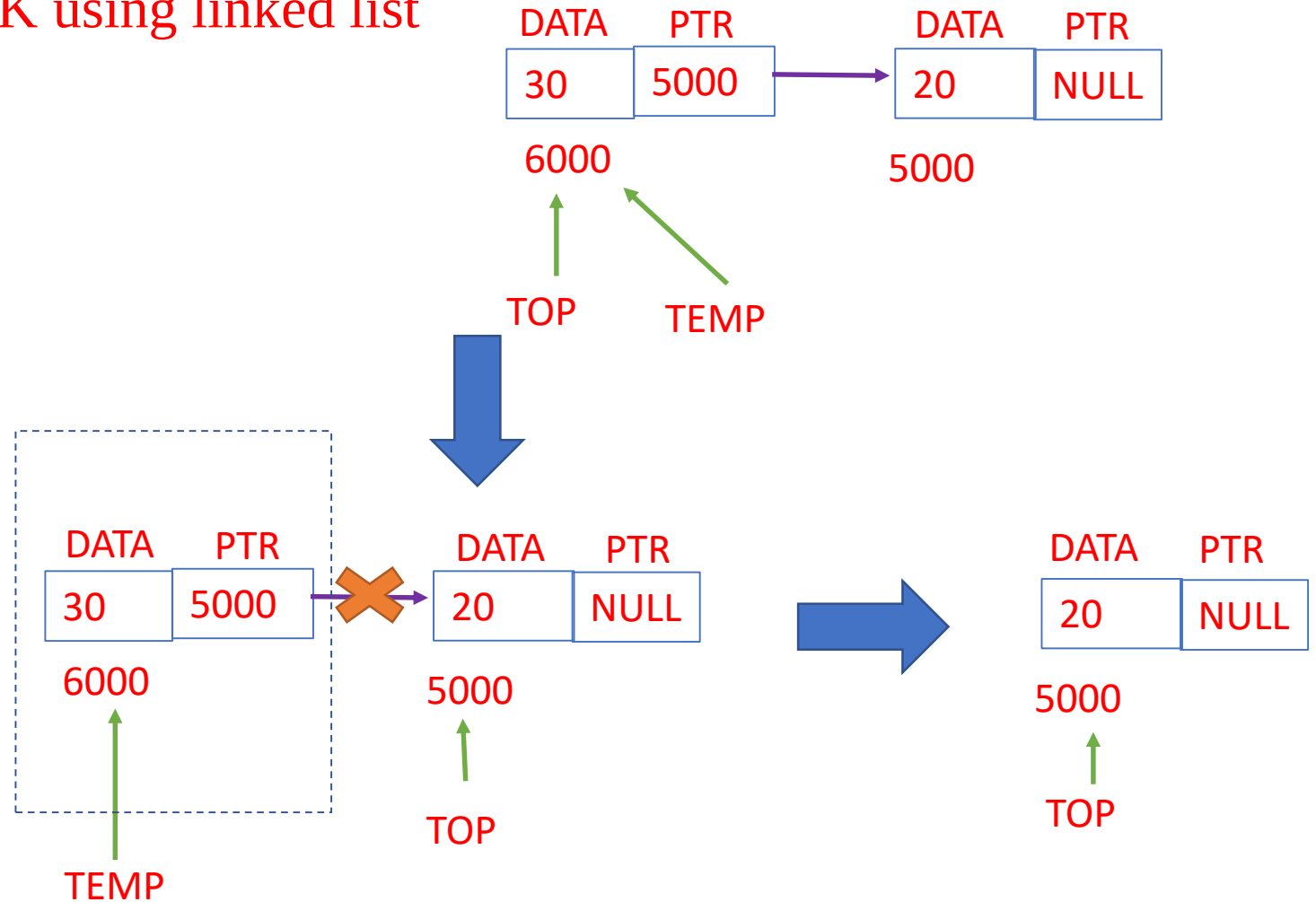
STEP-2: SET TEMP = TOP

TOP=TOP→ PTR

STEP-3: FREE(TEMP)

STEP-4: EXIT

2. SDELETE_LINK()



Delete an element in queue using linked list

STEP-1: If TOP = NULL

write “underflow”

go to step 4.

[End of if]

STEP-2: SET TEMP = TOP

TOP=TOP→ PTR

STEP-3: FREE(TEMP)

STEP-4: EXIT

3. SDELETE_LINK()

4. SDELETE_LINK()

